Economic Complexity — Written Exam (60 minutes)

Instructions

- Answer all questions in Part A, any 2 in Part B, and any 1 in Part C.
- Where relevant, reference concepts/authors from the readings and the lecture slides.
- Total: 30 points.

Part A — Short answers (5× 3 = 15 points; ~30 minutes)

A1. Define capabilities and explain how diversification and ubiquity in the country–product matrix—combined via the Method of Reflections—yield the Economic Complexity Index (ECI).

A2. Contrast the role of specialization with diversity in regional innovation.

A3. How can inter-industry labor flows be used to infer industry relatedness?

A4. What is a small-world in innovation networks? Explain how it can support knowledge diffusion and recombination simultaneously.

A5. How do social network segregation and urban topology jointly shape inequality?

Part B — Applied/data reasoning (choose any 2; 2 × 5 = 10 points; ~20 minutes)

B1. You advise a middle-income economy whose export basket is peripheral but has one near-core product. Propose a two-step diversification strategy (5–10 years) grounded in product proximity and capability accumulation. Specify at least one empirical indicator you would use.

B2. Design an empirical test to identify MAR vs. Jacobs effects using city–industry patents and co-inventor network metrics (e.g., clustering, path length). State: data, variables, identification idea, and expected coefficient signs.

B3. Propose a model to test whether geographic distance or network distance better predicts innovation diffusion across regions. List controls and one robustness check.

Part C — Integrative essay (choose any 1; 1 × 5 = 5 points; ~10 minutes)

C1. Discuss how capability accumulation, small-world structures, and technology-network topology jointly shape regional upgrading and shock resilience. Propose one feasible regional policy.

C2. Explain how firm-level revenue data for digital product trade complements product-space and urban externality insights for industrial policy in digitally transforming economies.

C3. Analyze the interplay between labor mobility, urban network segregation/topology, and the geography of innovation diffusion. What are the tensions for inclusive development?